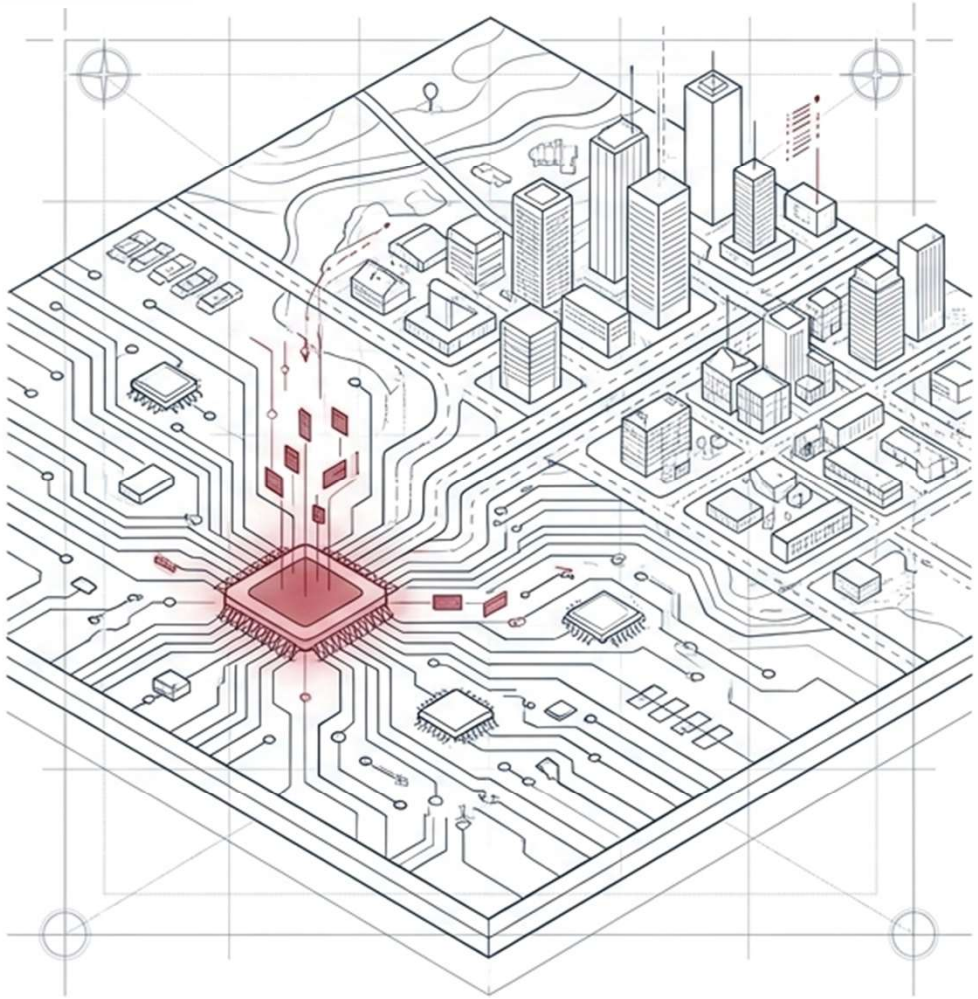
UNIVERSITÉ PARIS 1 PANTHÉON-SORBONNE

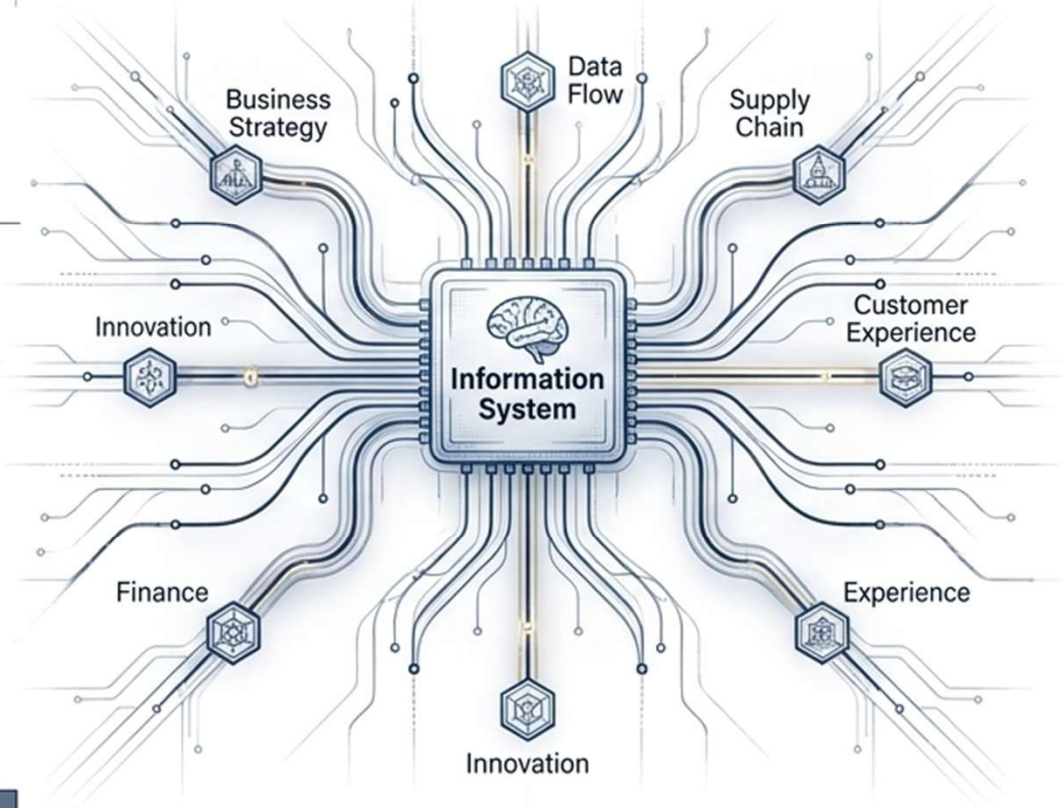
Information Systems Security: Anatomy of an Evolving Battlefield

From foundational architectures to the industrialization of cybercrime and AI-driven risks in 2026.



The Postulate: The IS as the Core of Value

Digital transformation has eradicated the "support function." Security is now a matter of organizational survival and competitiveness.



Operations

Business continuity and intellectual property **protection**.

Organization

Maintenance of **reputation** and partner trust.

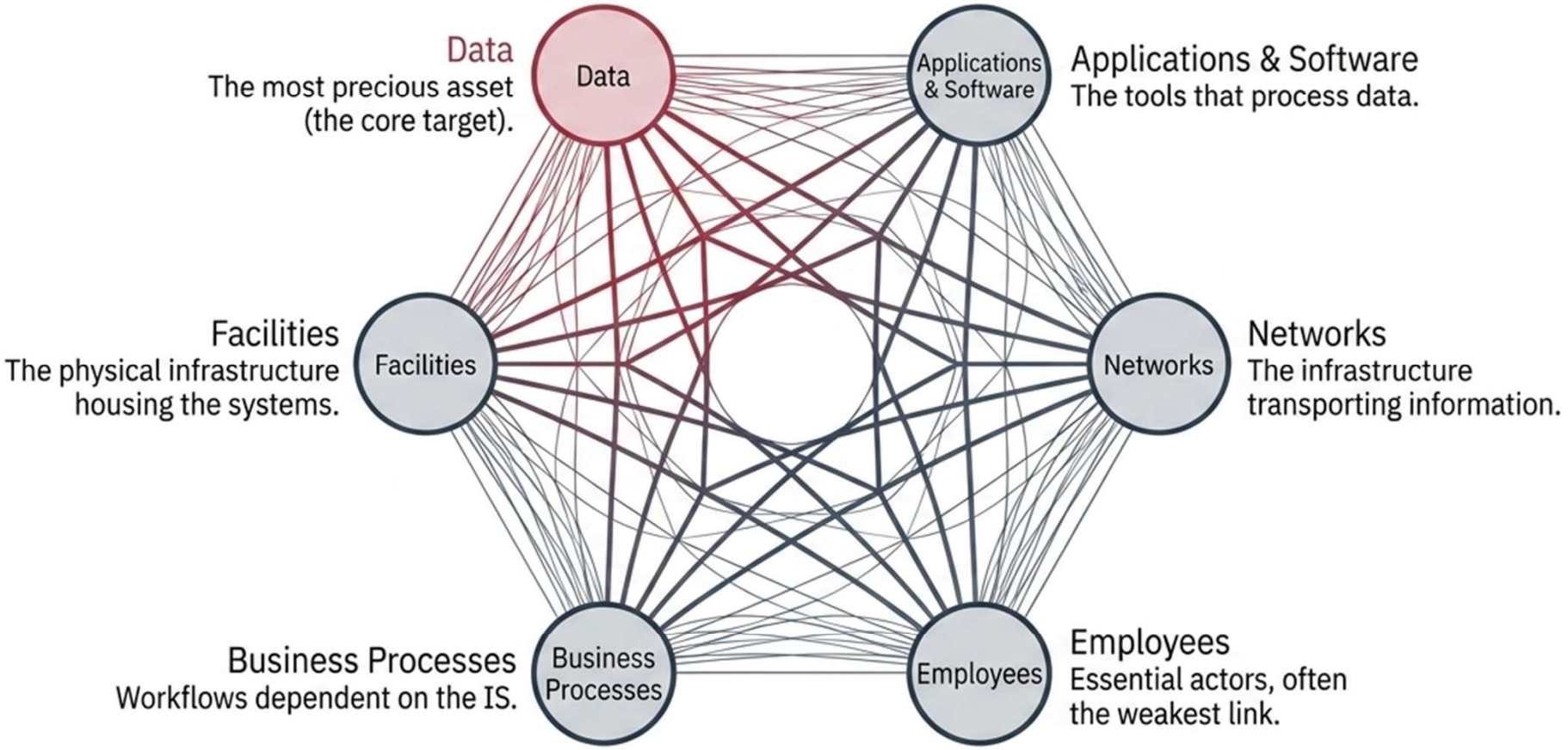
Employees

Data privacy and functional work environments.

Compliance

Legal mandates (e.g., NIS Directive, GDPR) and **liability**.

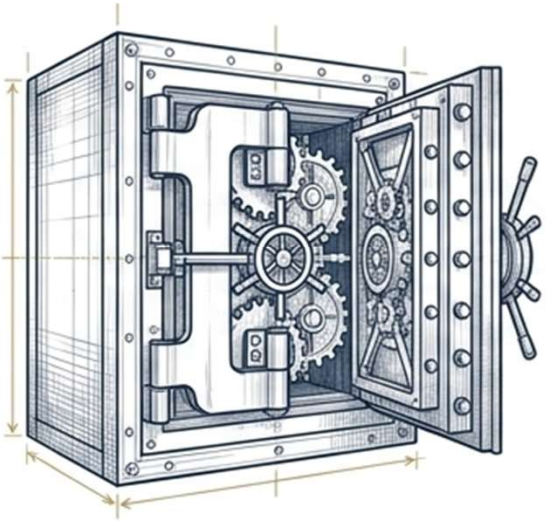
Defining the Territory: The 6 Components of the IS



Security must be holistic and coherent to cover this entire ecosystem.

Protecting the Crown Jewels: The Evaluation Criteria

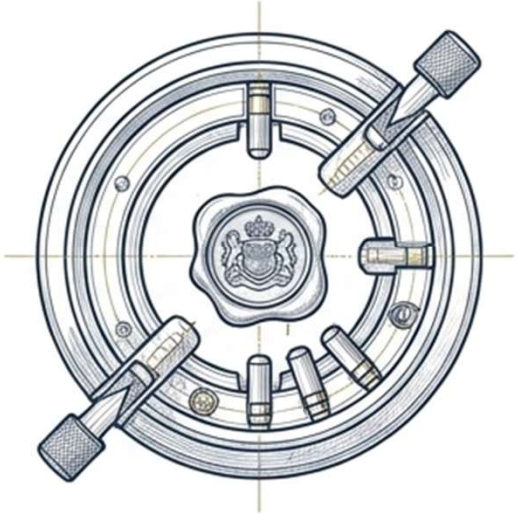
Confidentiality



Principle: Information is accessible only to authorized entities.

Risk: Unauthorized disclosure, corporate espionage.

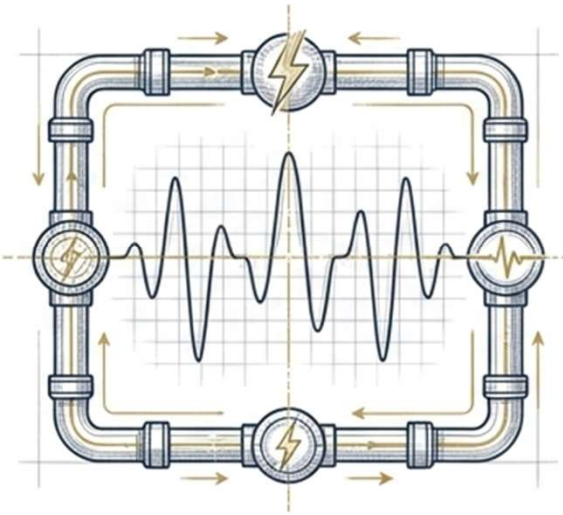
Integrity



Principle: Data is accurate, complete, and free from unauthorized modification.

Risk: Fraud, data manipulation, silent corruption.

Availability

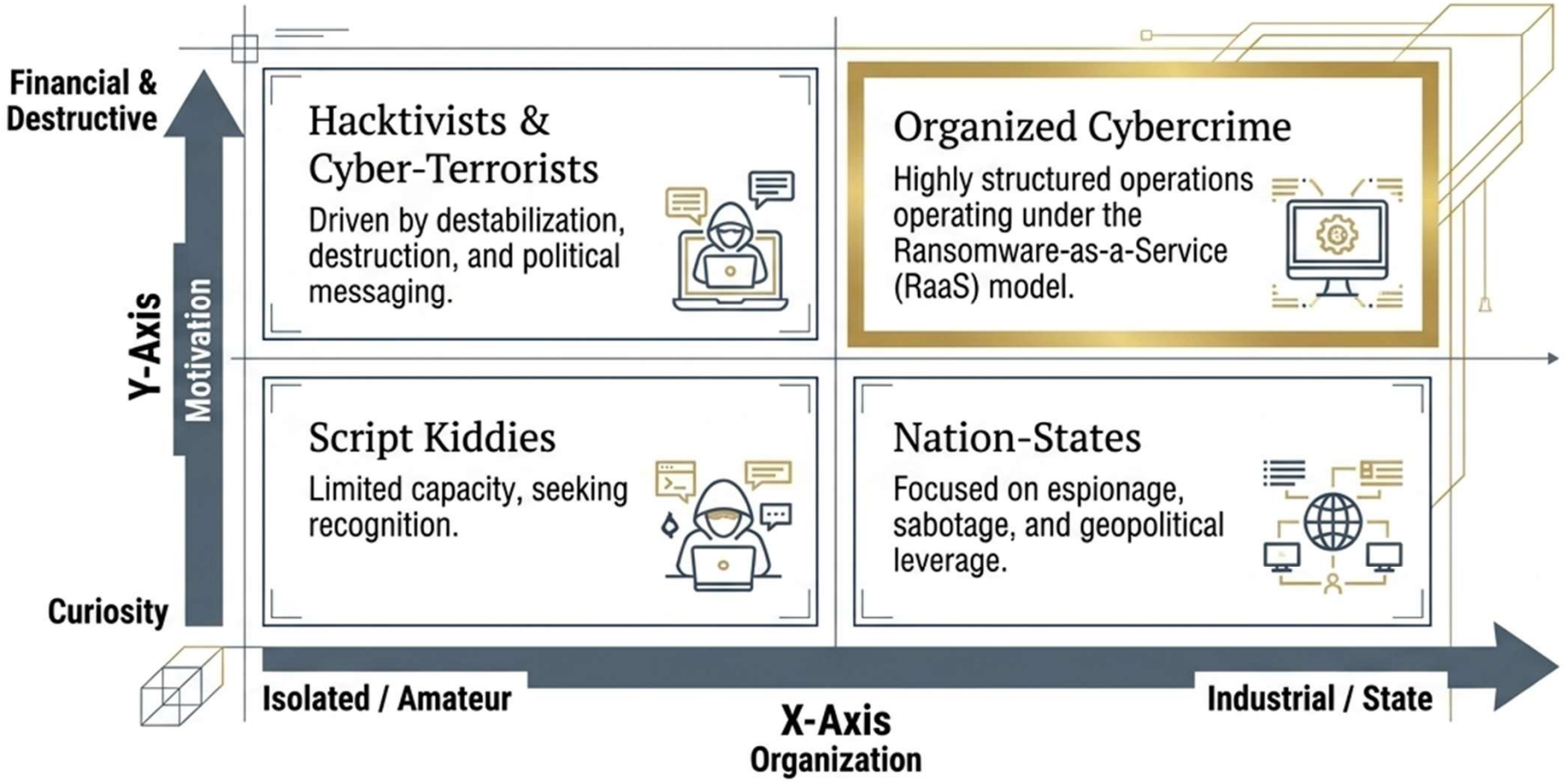


Principle: Systems and data are usable on demand by legitimate users.

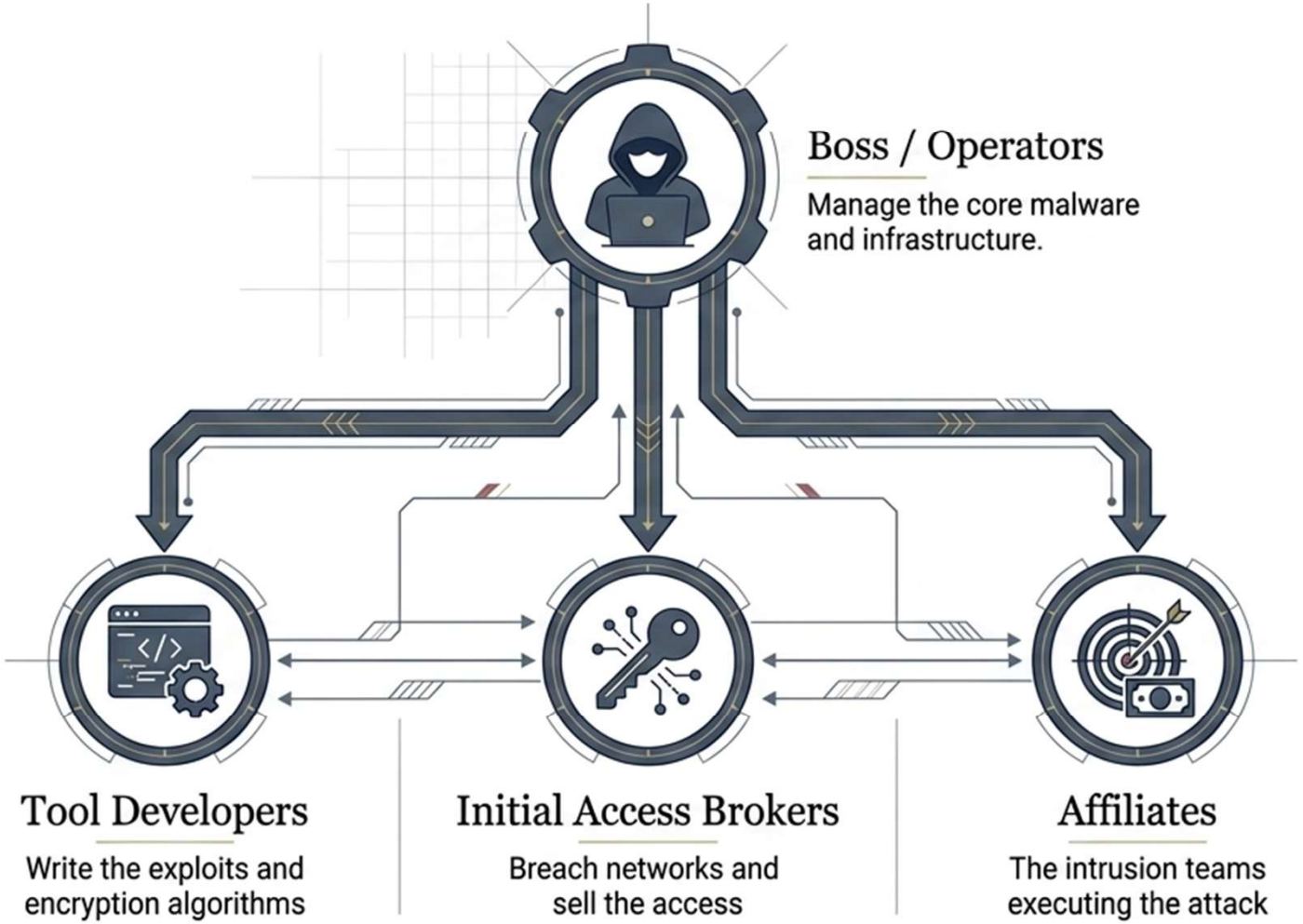
Risk: Business paralysis, Denial of Service (DoS).

The Threat Evolution Matrix

From isolated amateurs in the 1990s to the industrialized, structured adversaries of 2026.



The Industrialization of Extortion: Ransomware-as-a-Service (RaaS)



Data Sidebar (Chainalysis 2024/2025)

~$813.55M
Ransomware revenue (2024)

~$460M
Extorted in crypto by H1 2025

Geopolitical Link



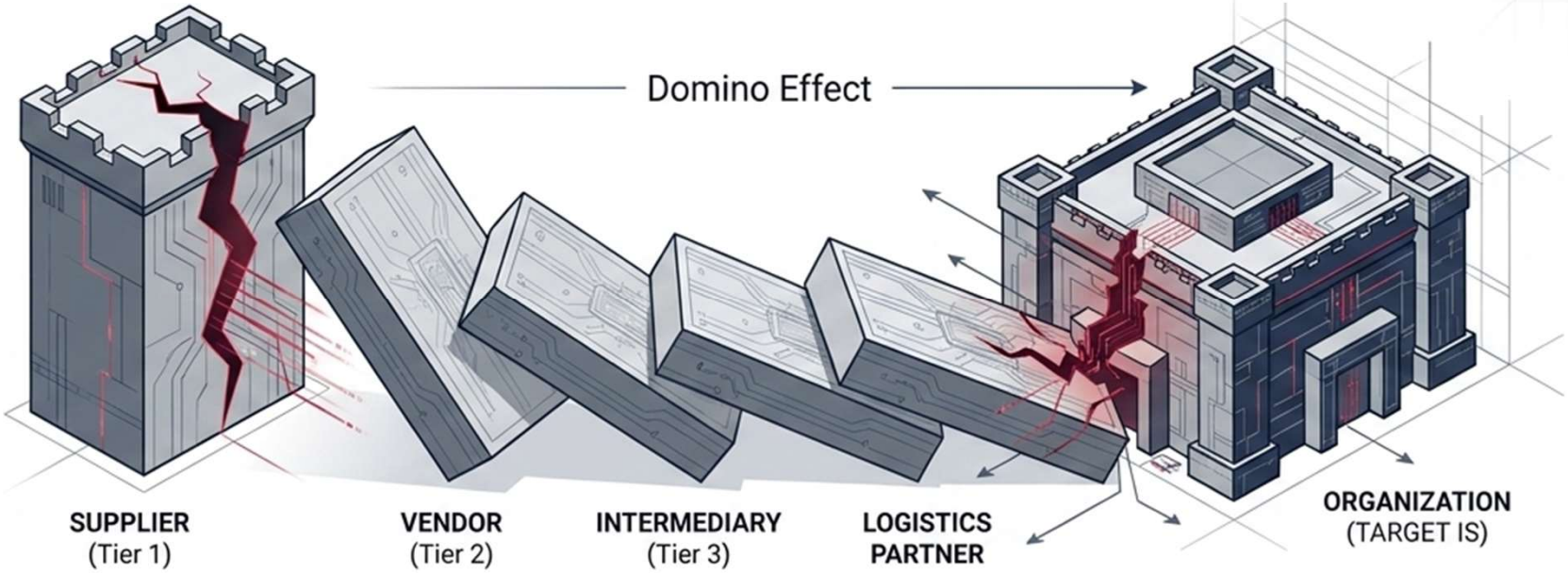
~74% of affiliate revenue traced to Russian jurisdictions

Strategic Dossier

Architectural Blueprint of a Digital Battlefield

Permeable Perimeters: The Supply Chain Vector

The weakest link is no longer inside your organization; it is embedded in your most trusted partnerships.



Software Vendors

Compromising a software update to infect all downstream users simultaneously.

Trusted Intermediaries

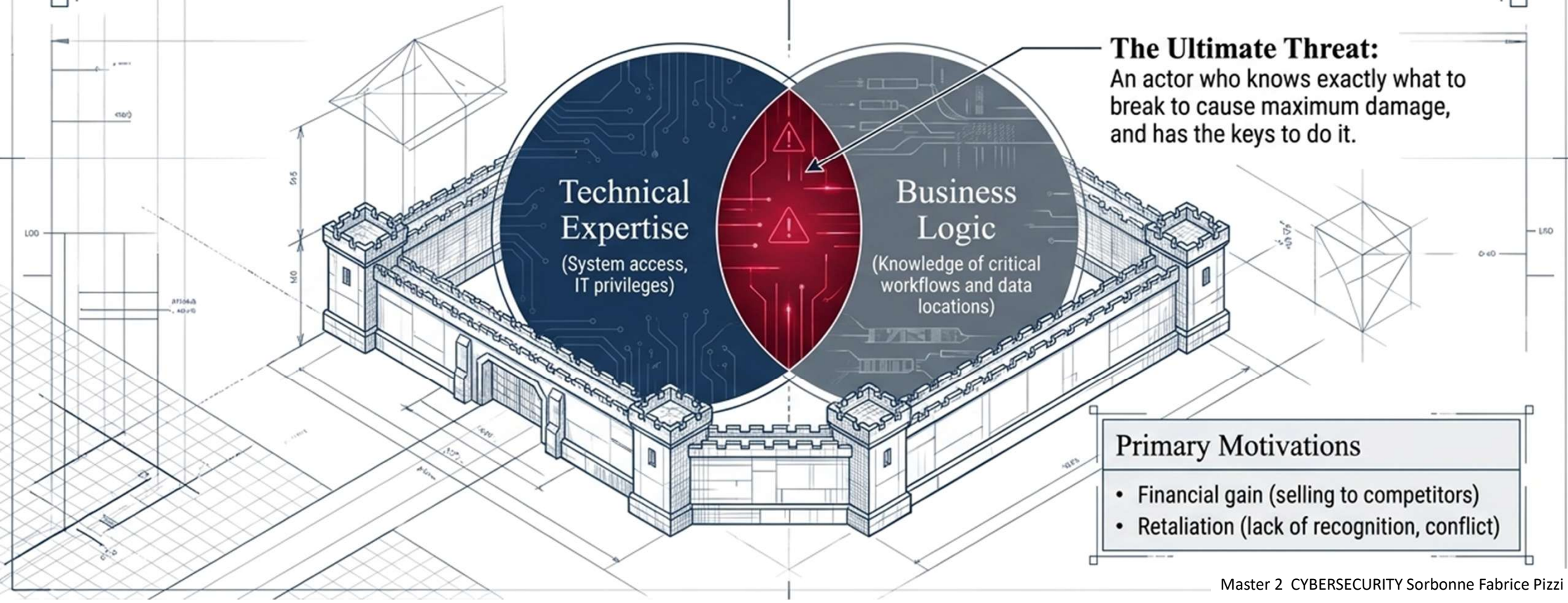
Targeting subcontractors or telecom operators who hold legitimate access to the target IS.

Mergers & Acquisitions

Acquiring a previously compromised company to gain a discreet, internal entry point.

The Insider Threat: Access Over Intent

While employees cause only 16% of attacks, their potential for critical disruption is absolute. Surveys indicate up to 50% of employees would consider corporate sabotage under certain conditions.



The Ultimate Threat:
An actor who knows exactly what to break to cause maximum damage, and has the keys to do it.

Primary Motivations

- Financial gain (selling to competitors)
- Retaliation (lack of recognition, conflict)

Information Warfare: E-Reputation and Destabilization

Not all attacks extract capital; many are designed to erode the foundation of citizen and market trust.

The Destabilization Playbook

DDoS (Distributed Denial of Service)

Paralyzing activity to demonstrate target weakness.

Data Leaks (Doxing)

Weaponizing sensitive information for blackmail or political pressure.

Defacement

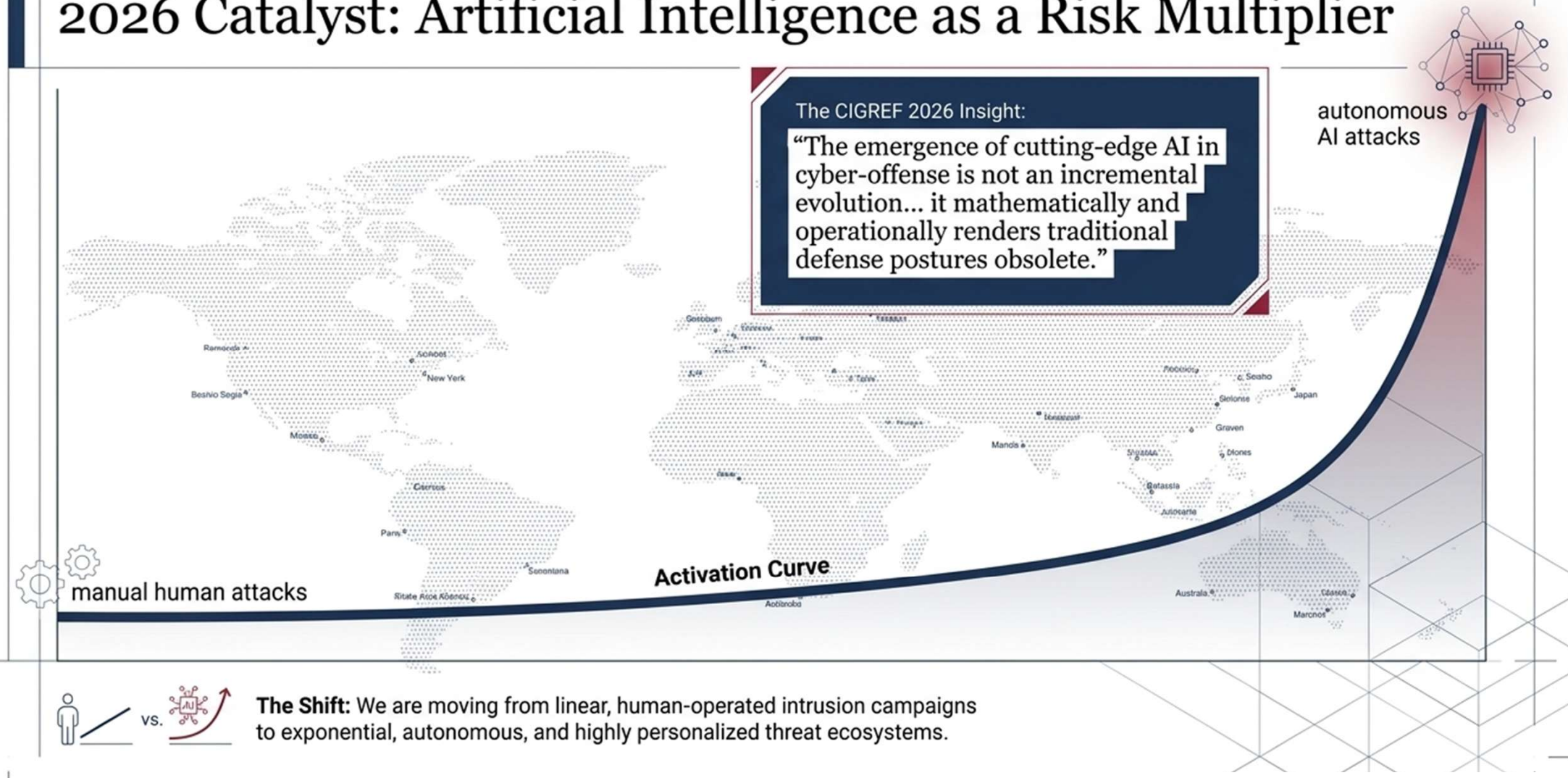
Altering public web presence for ideological messaging.

Synthetic Disinformation

Leveraging fake news to manipulate diplomacy, elections, and corporate valuations.



2026 Catalyst: Artificial Intelligence as a Risk Multiplier



AI Attack Vectors: Generative Deception and Excessive Agency

Path 1: Generative Deception (Targeting Trust)



Deepfakes & Audio Cloning

Usurping executive identities to bypass human verification.



Hyper-Personalized Spear-phishing

LLM-generated campaigns with perfect grammar and deep contextual knowledge of the target.

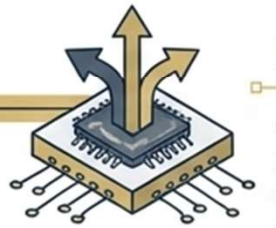
AI Risk Profile

Path 2: Autonomous Agents (Targeting Scale)



Automated Reconnaissance

AI scanning networks to identify vulnerabilities infinitely faster than human teams.

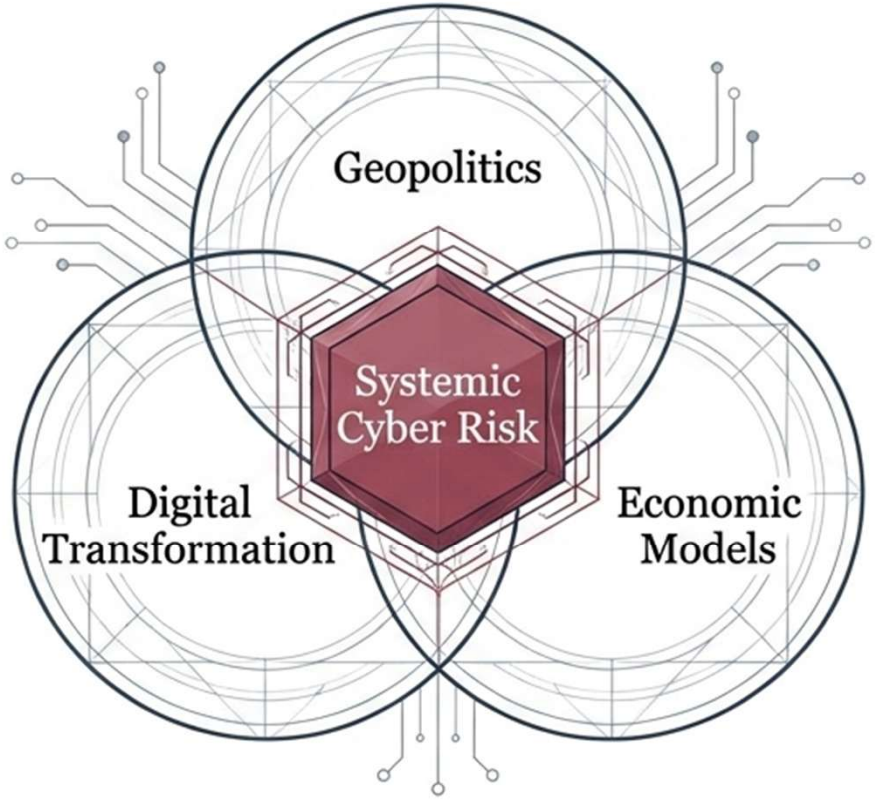


Lateral Movement

AI agents piloting tools, testing passwords, and pivoting through environments without requiring a human operator.

The EY 2026 Synthesis: A Structural Transformation of Risk

Cybersecurity has crossed a maturity threshold. It is no longer an isolated technical IT issue, but a systemic governance imperative.



The New Reality



Impacts are Transversal

Breaches simultaneously trigger operational collapse, financial hemorrhage, and reputational ruin.



Threats are Hybridized

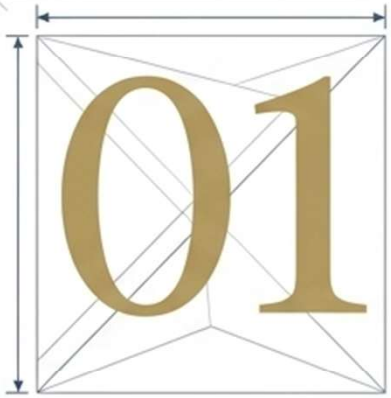
Industrialized crime syndicates now utilize state-level AI tools.



Required Response

Resilience requires an integrated approach combining executive governance, advanced AI defense technologies, and strict identity management.

Governing the Future: Four Axioms of Modern Security



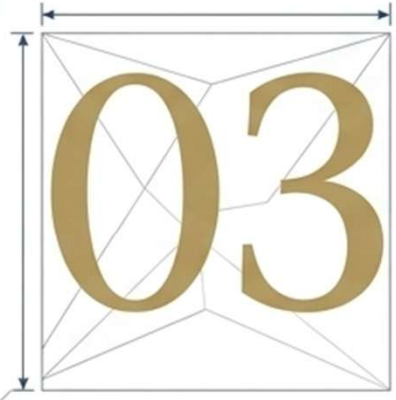
The IS is a Living Ecosystem

Protecting isolated technologies is futile; defense must account for data, people, processes, and facilities holistically.



The Adversary is Industrialized

We are defending against structured economies and automated agents, not rogue individuals.



Trust is the Primary Attack Surface

Vendors, partners, and elevated employee identities are the preferred entry vectors. Assume breach.



Security is Systemic Governance

The professional of tomorrow does not just secure networks; they govern relationships, permissions, dependencies, and autonomous algorithms.